Adobe Extension from Fiserv Payments

V1.4.1

Introduction

Carat is the global commerce platform from Fiserv that orchestrates payments and experiences for the world's largest businesses.

Commerce Hub is the modernized payment technology stack from Fiserv that facilitates the enablement of payments and value-added services for omni commerce. A connection to Commerce Hub will afford any merchant a single solution to meet their growing commerce needs.

Adobe Extension Overview

For the first time, an Adobe Commerce extension has been created for Carat from Fiserv Payments. This is a new initiative from Fiserv where we want to make Carat and Commerce Hub easily accessible to merchants across all categories and geographies.

Our intentions with this product are twofold:

1) We are building this extension to be fit for purpose.

   Fiserv will serve a merchant's needs as their business grows. At launch, you may just need domestic card processing, but then may require fraud tools to combat risk and chargebacks. Soon after, you desire to offer a better experience for your customers so tokenization and card-on-file will become a requirement. Your business expands into new regions. Fiserv can enable additional currencies and additional payment methods applicable to those markets.

   Wherever you are in your growth journey, Fiserv will fit your purpose wherever you are.

2) We want to make payments simple and fast.

   Fiserv has a strong legacy in payments and understands not only the complexity of payments, but also the regulatory landscape. Given our position in the market, we are on a mission to simplify payments for you so that you can accomplish your commerce goals at speed; hence, our motivation to develop our Adobe Commerce extension.

   While initial market entry of this will offer fundamental features, our commitment is to keep parity with the rich solutions that Carat offers. You can expect additional iterations soon.

Establish Account & Pricing

Please contact your sales representative or relationship manager.

Integration Guide
January 2026
v1.4.1

Business Use Cases

Current Business Use Cases

The supported business cases focus on enabling credit and debit card processing for card-not-present scenarios. At the point of purchase of purchase (e.g. online checkout) by the consumer, the extension facilitates the following.

Credit & Debit Cards

- Supports a consumer to present a credit or debit card to purchase goods and services as a “sale”.
- Supports a consumer to present a credit or debit card to purchase goods and services as an “authorization only” flow to be subsequently followed by one or more captures.
    - Supports void of a valid authorized transactions in cases where an order is canceled before fulfillment.
    - Supports full and multiple partial captures to manage complete fulfillment of an order, partial fulfillment, and split shipment
- Supports full and multiple partial refunds for returns of full and partial orders.
- Supports the opt-in or opt-out tokenization of cardholder data as a stored payment method for future use.
    - Supports the addition or deletion of store payment methods. This is a self-management tool given to the consumer within their account profile of the merchant storefront.
- Supports the input and capture of cardholder data through a PCI compliant and customizable implementation where sensitive data is hosted by Fiserv. This is rendered during consumer checkout via an iFrame provided by Commerce Hub’s Checkout solution.
- Supports split tender between credit / debit cards, gift cards, and Adobe store credit. A configurable maximum of 10 gift cards can be applied to a single order.
- Supports 3D Secure functionality in order to enhance payment card transaction security.

Fiserv Gift Cards

- Supports a consumer to perform a balance inquiry during checkout while applying a gift card as a form of payment. The balance will be applied to the shopping cart total.
- Supports a consumer to perform a balance inquiry only flow independent from checkout for those scenarios where a consumer is allowed to check their gift card balance.
- Supports a consumer to perform a gift card redemption
    - For the full amount of the purchase
    - For a partial amount of the purchase in combination with a credit/debit card and/or store credit
- Supports a consumer to present a gift card to purchase goods and services as a “sale”.
    - **NOTE: A gift card sale cannot be refunded, but only voided at this time.

- Supports a consumer to present a gift card to purchase goods and services as an "authorization only" flow to be subsequently followed by only one capture.
- Supports the ability for customers to apply multiple gift cards to a single order, up to a configurable maximum (default: 10).
- Supports voids for gift card authorizations.
    - Ability to input and capture gift card data through the same PCI compliant and customizable implementation as credit / debit cards fields are hosted by Fiserv. This is rendered during consumer checkout via an iFrame provided by Commerce Hub's Checkout Hosted Pages. While gift cards do not fall under PCI scope, this provides for a consistent integration and checkout experience for consumers.
- Enables multi-tender support – credit / debit card, gift card and store credit.
    - A configurable maximum of 10 gift cards can be applied to a single order.
    - In a multi-tender scenario, gift cards are always authorized before authorizing a card payment method. If the subsequent card auth is declined, the extension will automatically void the gift card authorization. If a sale is configured, a merchant must then refund the amount based on their Standard Operating Procedures.

Features and Functionality

Countries & Currencies

- United States (domiciled merchants)
- USD

Enablement of additional countries will be enabled over time. Please note that with this current extension, merchants can accept purchases from other countries, but the prices and amounts will be presented in USD. Any transactions purchased from abroad will incur foreign exchange fees which will be borne by the cardholder.

Payment Methods

- Visa
- Mastercard
- American Express
- Discover
- Diners Club
- JCB
- Fiserv Gift Cards

Payment Functionality

- Authorization only
  This is a function to reserve a defined amount on the consumer's payment method with the intention of sending a subsequent capture request to fully charge and settle funds.

- Sale (authorization + capture)
  This is a function that combines authorization and capture together in a single operation. This will remove the need to send a subsequent capture call and normally used for products that are immediately delivered to the consumer – access to content, downloadable software, etc.

- Void
  This function voids (e.g. cancels) valid authorizations that have been secured on a payment method. This returns the reserved amount back to the consumer payment method. This can be executed so long as a capture has not been initiated. This is typically used for scenarios where an order needs to be canceled.

- Capture
  - Full
  - Multiple Partial

- Refund
  - Full
  - Multiple Partial

  **NOTE. Gift card refunds are not supported. Most merchants have a refund policy that returns funds to either store credit or a different gift card.

- Tokenization
  This enables a form of payment to be tokenized and stored in the consumer's profile for card-on-file use cases. When enabled, the consumer can opt to store their card for future use.
  NOTE: This is only available for credit and debit cards – not gift cards.

- Fiserv Gift Cards
  - Balance Inquiry
  - Redemption
    - The redemption function has been built to follow the traditional card flows for auth and capture to be consistent with customary process flows. A sales flow can be configured as well where an authorization and capture can be executed in a single request.
    - Customers can apply multiple gift cards to a single order, up to a configurable maximum (default: 10).
    - Split Payment orchestration is possible for customers so that they can split order payment between gift, payment cards and Adobe Store Credit

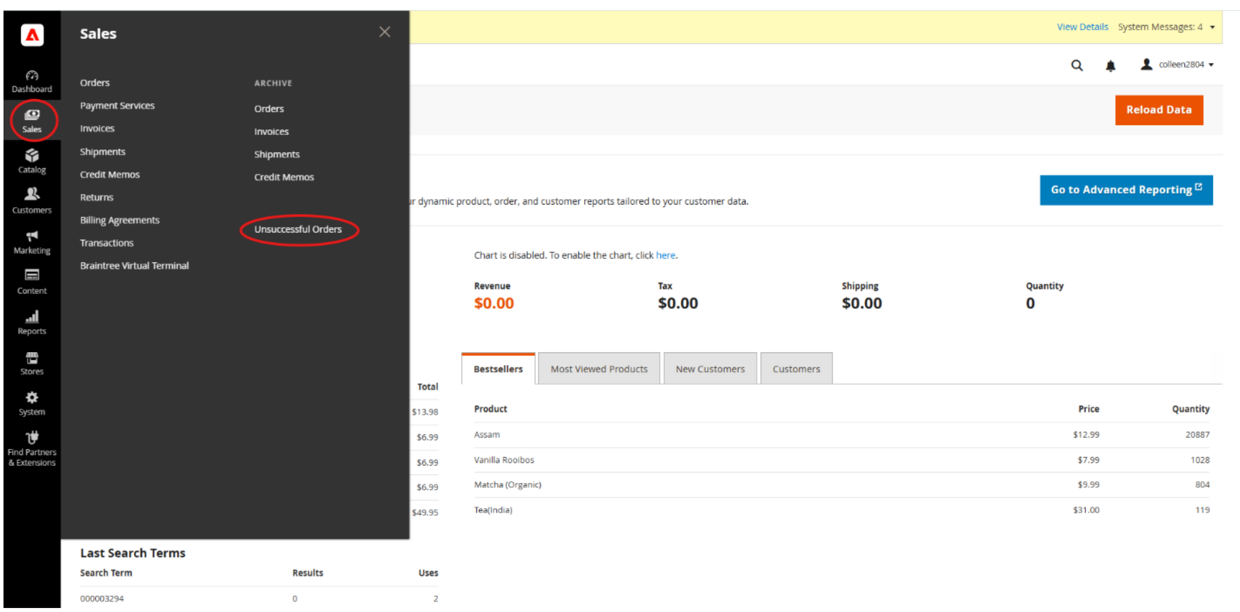
Security & PCI Compliance

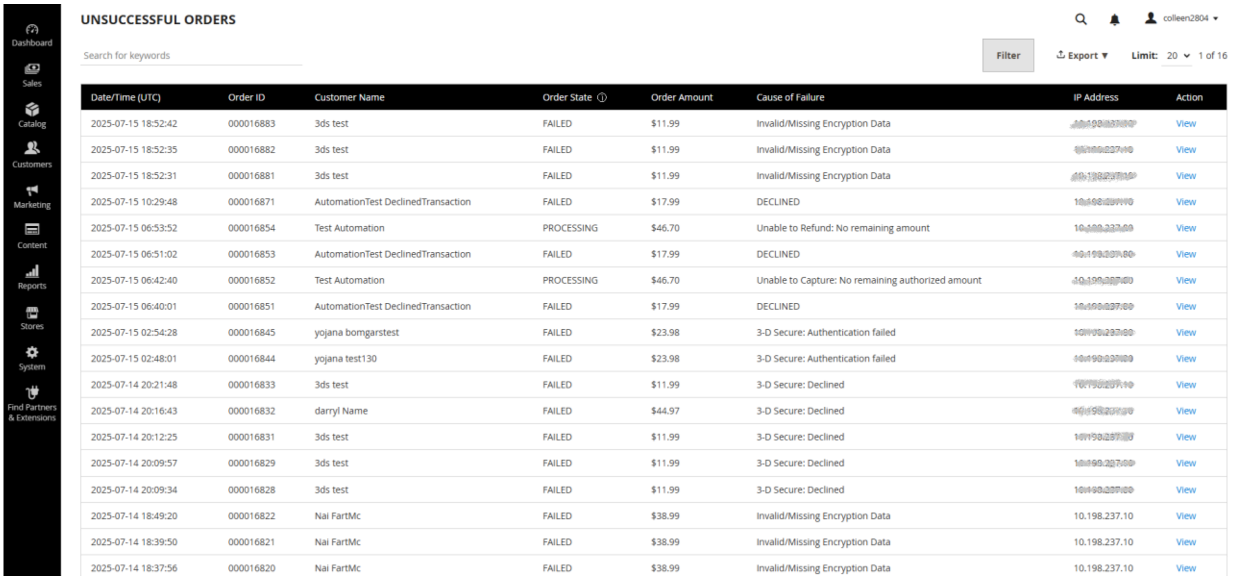
- PCI Compliance
  - Fiserv understands the importance of mitigating PCI risk for merchants and have taken liberty to leverage Commerce Hub’s Checkout (formerly known as Secure Data Capture). The plugin enables the use of an SAQ A iFrame integration to facilitate the collection of sensitive card information as well as offering field customization to better match merchant website branding.
- 3D Secure
  - Support for 3D Secure has been added to enhance both payment card and stored payment card transaction security.

Enhancements to Adobe Commerce native functionality

In order to offer the best in-class payment plugin to our Adobe Commerce clients, some enhancements and additional features were built within the Adobe Commerce framework. Given our position in the market, we are on a mission to simplify payments for merchants so that they can accomplish their commerce goals at speed.

- Unsuccessful Order Dashboard
  - A new dashboard has been introduced to provide detailed, exportable insights into orders with unsuccessful transactions, including cause of failure, payment gateway response, and other relevant information.





Date/Time (UTC)	Order ID	Customer Name	Order State	Order Amount	Cause of Failure	IP Address	Action
2025-07-15 18:52:42	000016883	3ds test	FAILED	$11.99	Invalid/Missing Encryption Data	[illegible]	View
2025-07-15 18:52:35	000016882	3ds test	FAILED	$11.99	Invalid/Missing Encryption Data	[illegible]	View
2025-07-15 18:52:31	000016881	3ds test	FAILED	$11.99	Invalid/Missing Encryption Data	[illegible]	View
2025-07-15 10:29:48	000016871	AutomationTest DeclinedTransaction	FAILED	$17.99	DECLINED	[illegible]	View
2025-07-15 06:53:52	000016854	Test Automation	PROCESSING	$46.70	Unable to Refund: No remaining amount	[illegible]	View
2025-07-15 06:51:02	000016853	AutomationTest DeclinedTransaction	FAILED	$17.99	DECLINED	[illegible]	View
2025-07-15 06:42:40	000016852	Test Automation	PROCESSING	$46.70	Unable to Capture: No remaining authorized amount	[illegible]	View
2025-07-15 06:40:01	000016851	AutomationTest DeclinedTransaction	FAILED	$17.99	DECLINED	[illegible]	View
2025-07-15 02:54:28	000016845	yojana bomgarstest	FAILED	$23.98	3-D Secure: Authentication failed	[illegible]	View
2025-07-15 02:48:01	000016844	yojana test130	FAILED	$23.98	3-D Secure: Authentication failed	[illegible]	View
2025-07-14 20:21:48	000016833	3ds test	FAILED	$11.99	3-D Secure: Declined	[illegible]	View
2025-07-14 20:16:43	000016832	darryl Name	FAILED	$44.97	3-D Secure: Declined	[illegible]	View
2025-07-14 20:12:25	000016831	3ds test	FAILED	$11.99	3-D Secure: Declined	[illegible]	View
2025-07-14 20:09:57	000016829	3ds test	FAILED	$11.99	3-D Secure: Declined	[illegible]	View
2025-07-14 20:09:34	000016828	3ds test	FAILED	$11.99	3-D Secure: Declined	[illegible]	View
2025-07-14 18:49:20	000016822	Nai FartMc	FAILED	$38.99	Invalid/Missing Encryption Data	10.198.237.10	View
2025-07-14 18:39:50	000016821	Nai FartMc	FAILED	$38.99	Invalid/Missing Encryption Data	10.198.237.10	View
2025-07-14 18:37:56	000016820	Nai FartMc	FAILED	$38.99	Invalid/Missing Encryption Data	10.198.237.10	View

- Configurable Invoice Creation for Gift Card Transactions
    - Merchants can choose to generate invoices upon authorization or fulfillment for those orders that are covered entirely by gift card transactions.

Adobe Commerce Platform Compatibility

Edition:
- Adobe Commerce (cloud)
- Adobe Commerce (on-prem)
- Magento Open Source

Adobe Commerce Store Version:
- v2.4.7-p1
- v2.4.8-p1

Fiserv Checkout SDK
- Version 3.7.16

Browers Tested (Storefront and Admin):
- Chrome
- Firefox
- Microsoft Edge
- Safari

Quick Start

The first order of business is to install, configure, and run your first transaction in our test (e.g. CERT) environment. The extension is built ready-to-go. The installation has been constructed to have pre-configured settings to run a transaction within minutes upon configuring your test account and credential information.

Test Account and Credentials

To run your first transaction, you need to establish a test account and credentials through Dev Studio. If this has not been established previously, follow these quick instructions.

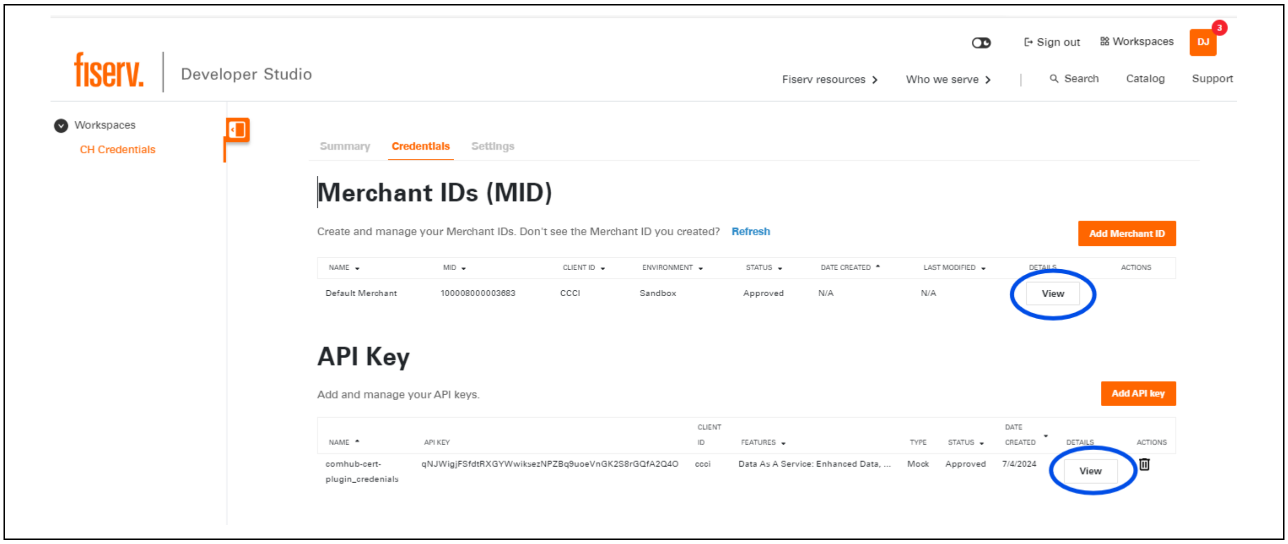
1) Navigate to the Fiserv Developer Studio (Dev Studio-- https://developer.fiserv.com/)
2) Create an account / login
3) Create a workspace
    a. Create a Merchant ID and Termina ID (e.g. MID and TID)
    b. Create API Key and API Secret

All information about how to execute the above can be found at:
https://developer.fiserv.com/support/docs/?path=docs/guides/workspaces.md

As a result of these steps, you will have access to the following:

- Merchant ID
- Terminal ID
- API Key
- API Secret

You will need these four values to run your first transaction. Please save them in a convenient location. They can be found in the Credentials tab of your Dev Studio Workspace.


fiserv.
Developer Studio
Sign out
Workspaces
Fiserv resources
Who we serve
Search
Catalog
Support
Workspaces
CH Credentials
Summary
Credentials
Settings
Merchant IDs (MID)
Create and manage your Merchant IDs. Don't see the Merchant ID you created? Refresh
Add Merchant ID
Default Merchant 100008000003683 CCCI Sandbox Approved N/A N/A
View
API Key
Add and manage your API keys.
Add API key
comhub-cert-plugin_credenials
ccci
Data As A Service: Enhanced Data, ...
Mock
Approved
7/4/2024
View

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January 2026
v1.4.1

Integration

Integration Playbook

As a guide to an efficient integration, we suggest the following actions and their order of operations.

1) Begin securing a LIVE PRODUCTION account – e.g. MID and TID. Please contact sales or if you are an existing client, your relationship manager
    a. This step will start the any commercial and business requirements – including pricing, contracting, and onboarding
    b. CERT accounts are automatically provisioned through Dev Studio
2) Secure CERT accounts and credentials through Dev Studio
3) Install Adobe extension from Fiserv Payments
4) Configure Adobe extension
5) Run first transaction as a proof of concept that payment extension is working
6) Configure extension for your business requirements
7) Implement and Test
    a. Consumer checkout. Integrate extension into your storefront process flows.
    b. Admin interface. Integration into Adobe admin dashboard.
    c. Consumer payment method management. Optional – if enabled.
8) Field customizations
9) UAT
10) End to End
11) Go Live

Step 1: Download & Installation

There are two options to access and download the Adobe extension from Fiserv Payments.

Option 1: Install via Composer (recommended)

Add the Fiserv repository using the following command line:

```
config repositories.fiserv_payments git https://
github.com/Fiserv/adobe-commerce-plugin.git
```

Require Fiserv Payments

```
composer require fiserv/payments:dev-master#v0.0.5
```

Option 2: Install compressed library (e.g. .tar / .zip file)

Extract compressed plugin to the following directory:

```
<Adobe Commerce Root Directory>/app/code/Fiserv
```

Upon downloading with either Option 1 or 2, please follow the below steps to enable the extension:

1) Enable Fiserv Payments module
```
php bin/magento module:enable Fiserv_Payments --clear-static-content
```

2) Update modules
```
php bin/magento setup:upgrade
```

3) Deploy static content
```
php bin/magento setup:static-content:deploy –f
```

4) Clear cache
```
php bin/magento c:c
php bin/magento c:f
```

5) For security reasons, Fiserv requires that all domains for development, UAT, production, etc.) must be whitelisted. Whitelisting can be facilitated through your sales or account representative.

**NOTE: Fiserv Gift Cards have been added as a form of payment. To support, the installation will include the following schema updates.

```
* New table: valuelink_transaction

* Altered table: quote
     * Added column valuelink_cards
```

```
    * Added column valuelink_cards_amount
    * Added column base_valuelink_cards_amount

* Altered table: quote_address
    * Added column valuelink_cards
    * Added column valuelink_cards_amount
    * Added column base_valuelink_cards_amount
```

Step 2: Fiserv Payments Configuration

!!! REMINDER: Configuration and first transaction cannot be completed until a test account and credentials have been created. See Quick Start section (above).

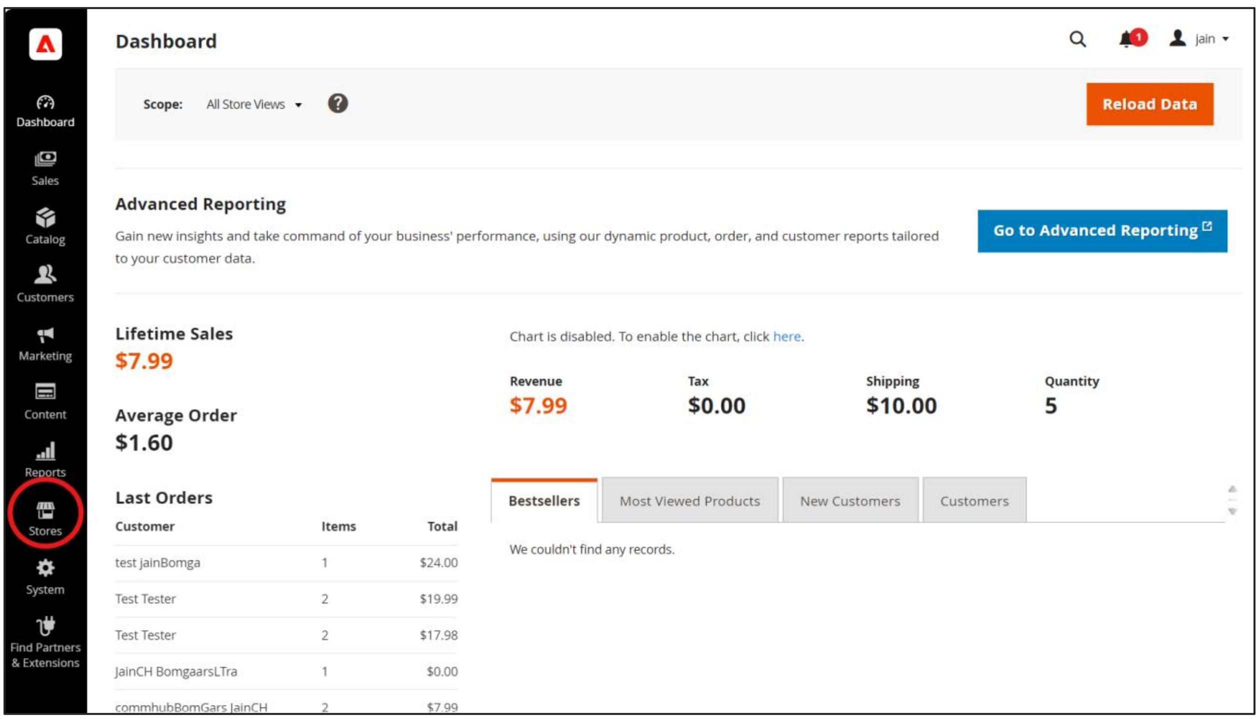
Upon downloading and installing the Adobe extension from Fiserv Payments, the next step is to configure the plugin so that we can run a first test transaction. This will ensure that all initial environmental and configuration settings are in place.

Below are steps to navigate to the Fiserv Payments configuration. Within the Adobe Commerce Administrative panel, navigate to:

Stores > Configuration > Sales > Payment Methods > Fiserv Payments

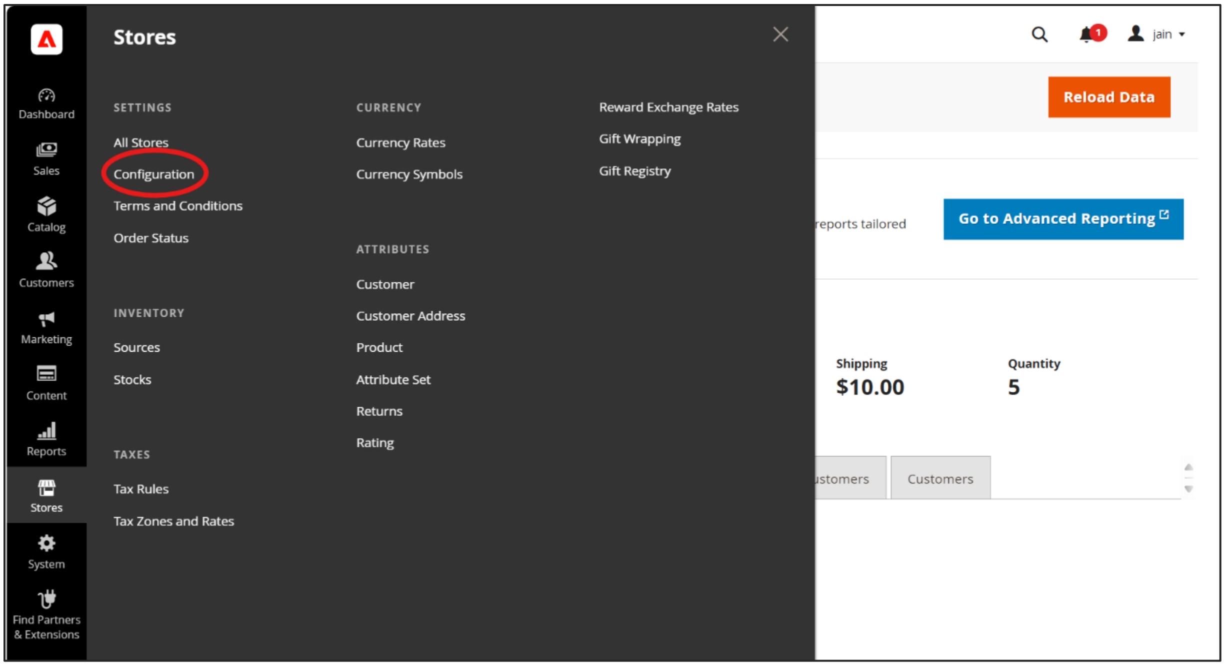
Navigate to Stores

Stores > Configuration > Sales > Payment Methods > Fiserv Payments



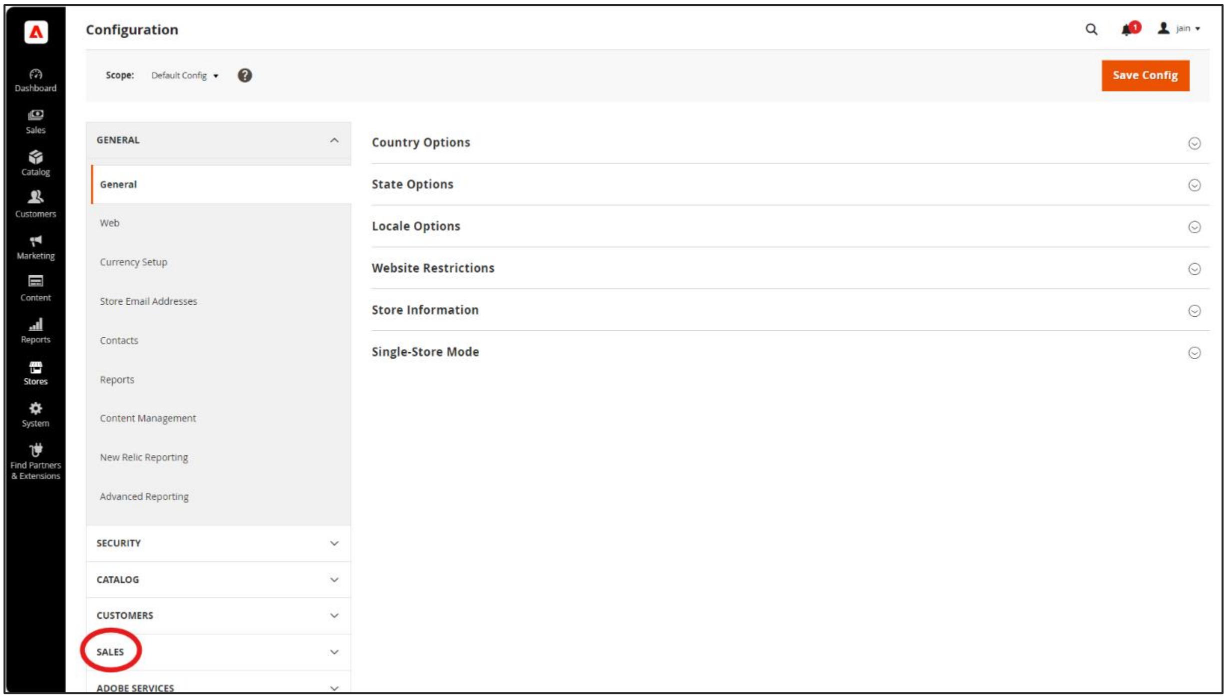
Navigate to Configuration

Stores > Configuration > Sales > Payment Methods > Fiserv Payments



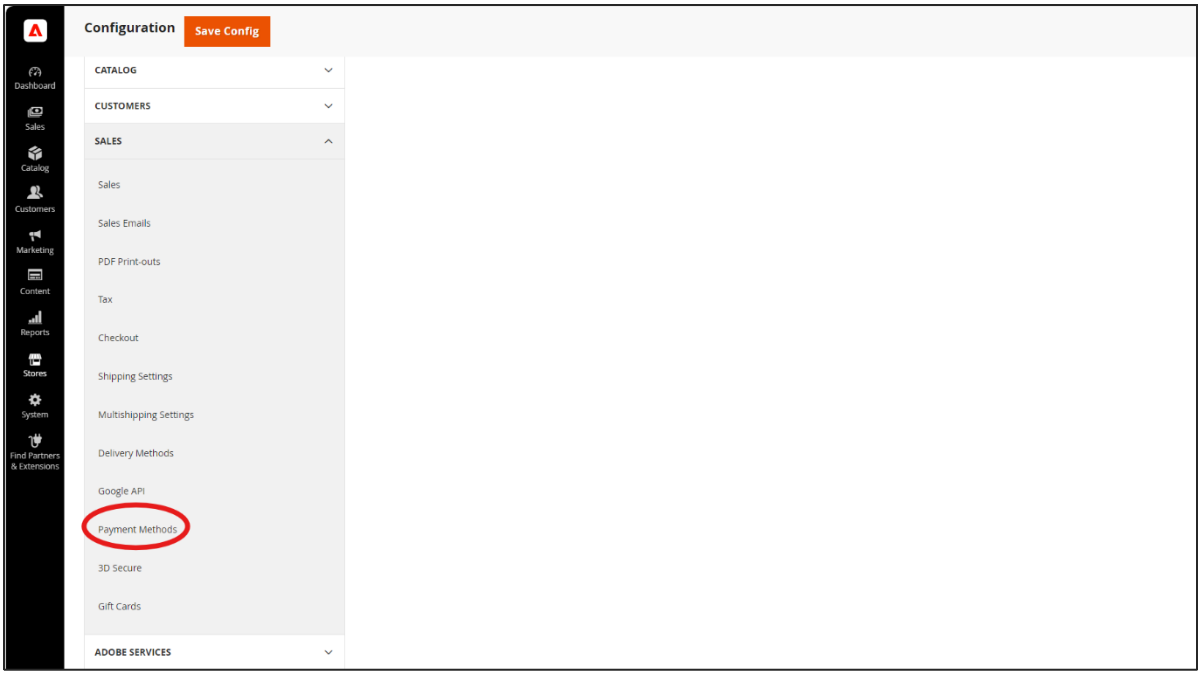
Navigate to Sales

Stores > Configuration > Sales > Payment Methods > Fiserv Payments


Configuration
Dashboard
Sales
Catalog
Customers
Marketing
Content
Reports
Stores
System
Find Partners & Extensions
Scope: Default Config
Save Config
GENERAL
General
Web
Currency Setup
Store Email Addresses
Contacts
Reports
Content Management
New Relic Reporting
Advanced Reporting
SECURITY
CATALOG
CUSTOMERS
SALES
ADOBE SERVICES
Country Options
State Options
Locale Options
Website Restrictions
Store Information
Single-Store Mode

Navigate to Payment Methods

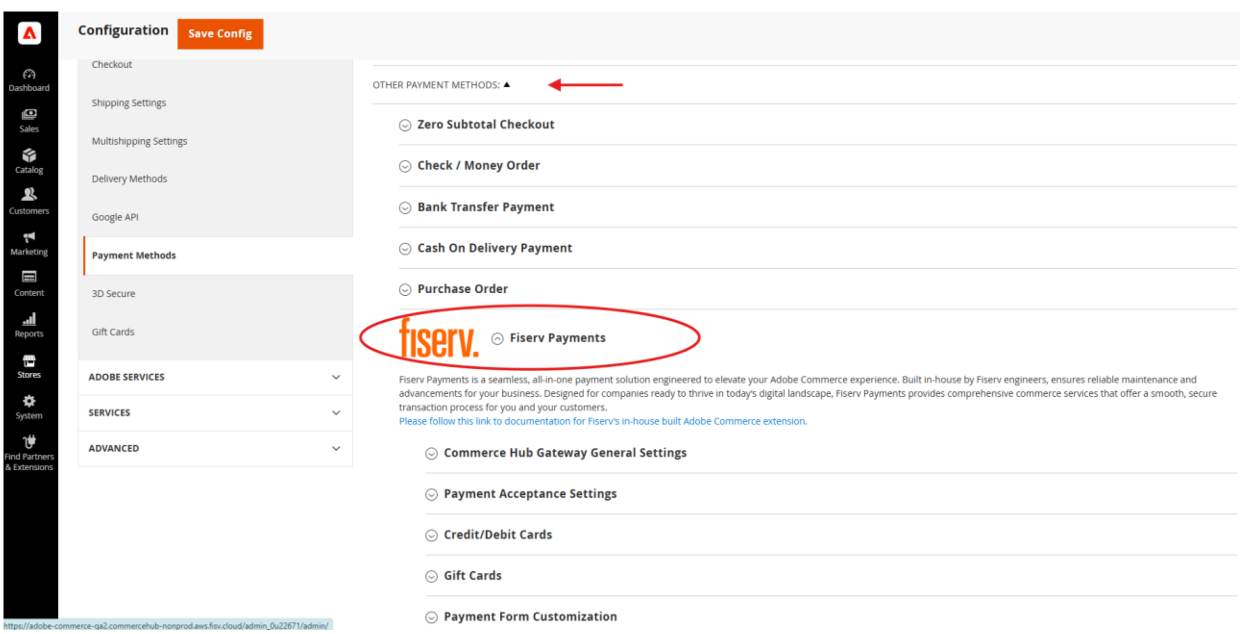
Stores > Configuration > Sales > **Payment Methods** > Fiserv Payments



Navigate to Fiserv Payments

Stores > Configuration > Sales > Payment Methods > **Fiserv Payments**

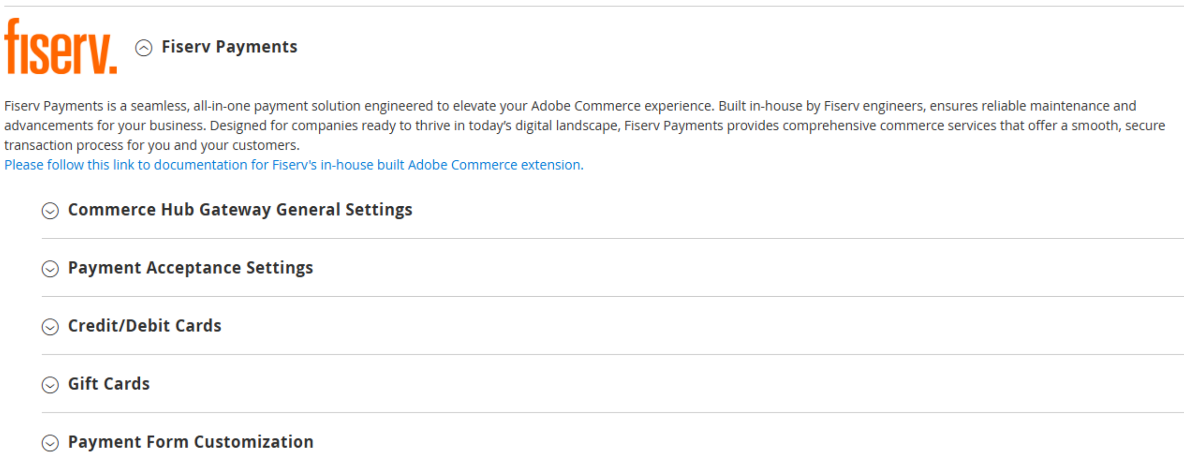
**NOTE: Fiserv Payments will be listed under "OTHER PAYMENT METHODS"



Expand Commerce Hub Gateway Settings

Stores > Configuration > Sales > Payment Methods > Fiserv Payments > Commerce Hub Gateway General Settings

Expand the “Fiserv Payments” payment method to expose “Commerce Hub Gateway General Settings”



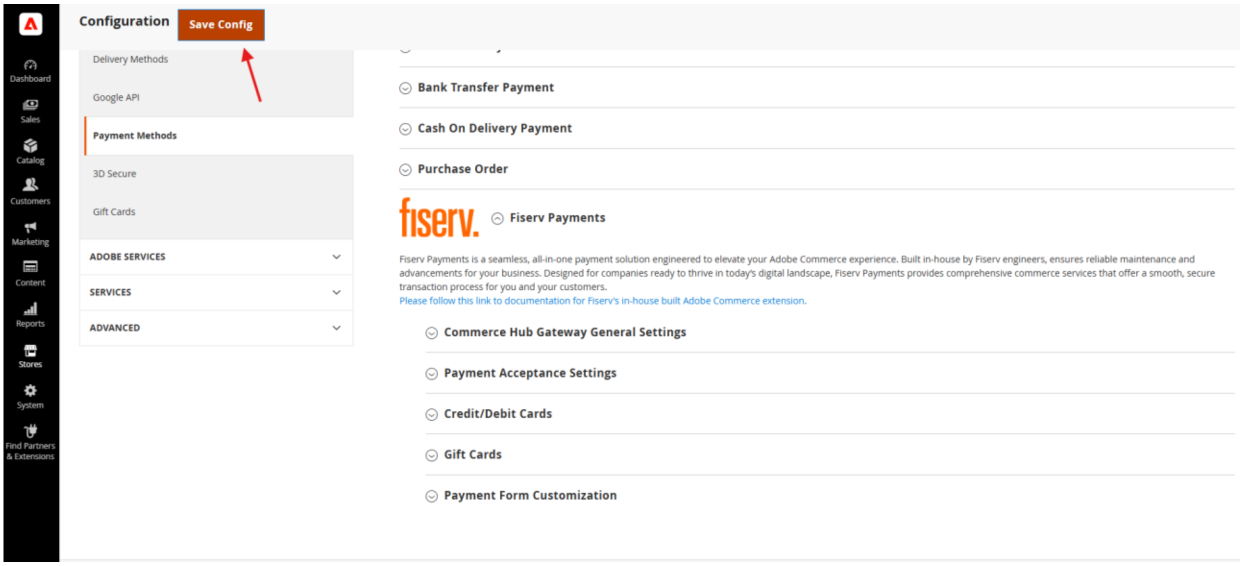
Upon expansion, a list of configuration areas will be revealed. All configurations will be done on this page. The configuration section has been categorized the following:

- Commerce Hub General Settings
- Payment Acceptance Settings
- Credit/Debit Cards
- Gift Cards
- Payment form Customization

Details of these categories are explained in Step 3.

Saving Configurations & Cache Management

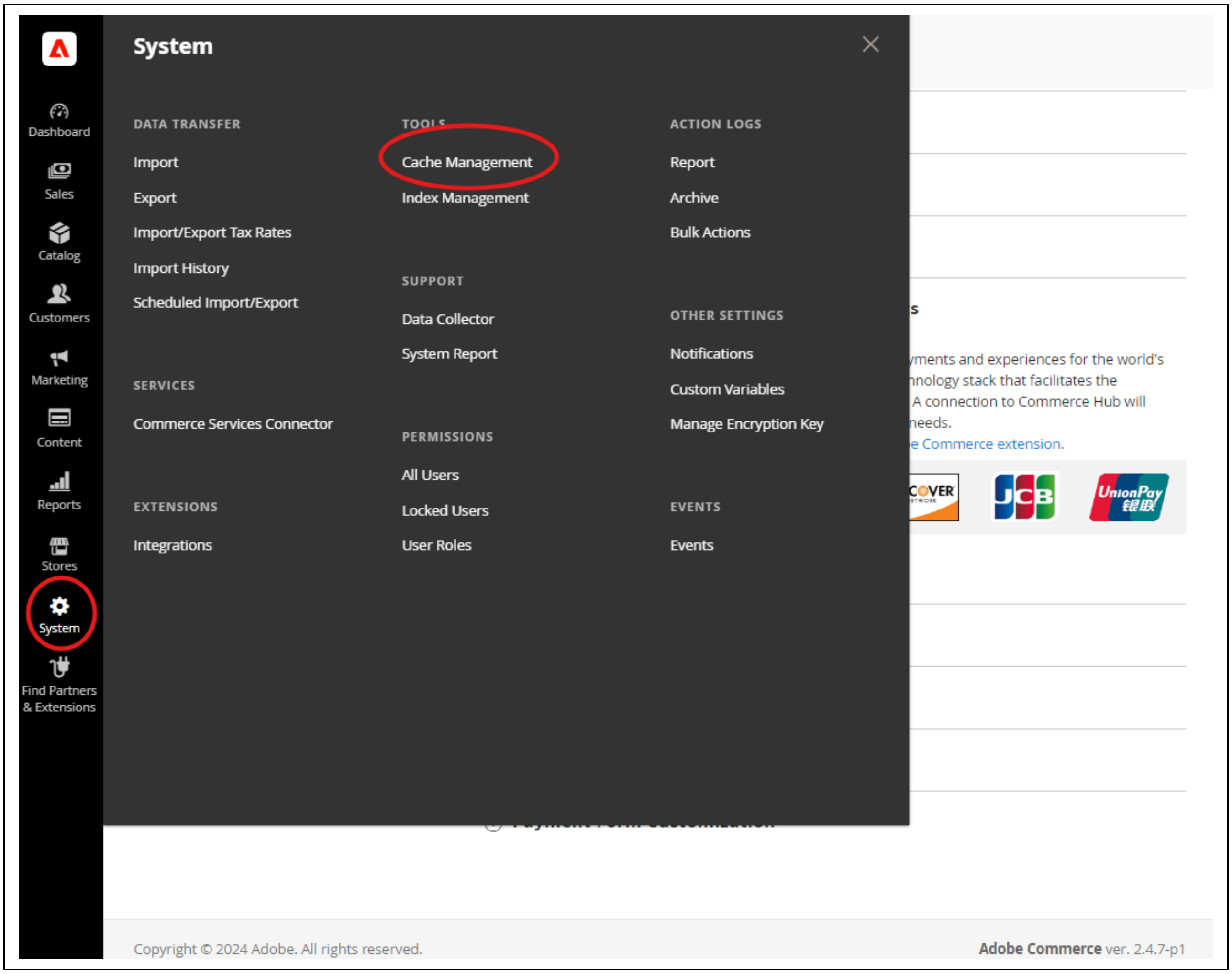
Each time a configuration change is made, one must remember to save changes. To do this, you must choose "Save Config" located at the top of the page. This must be completed for changes to take effect.



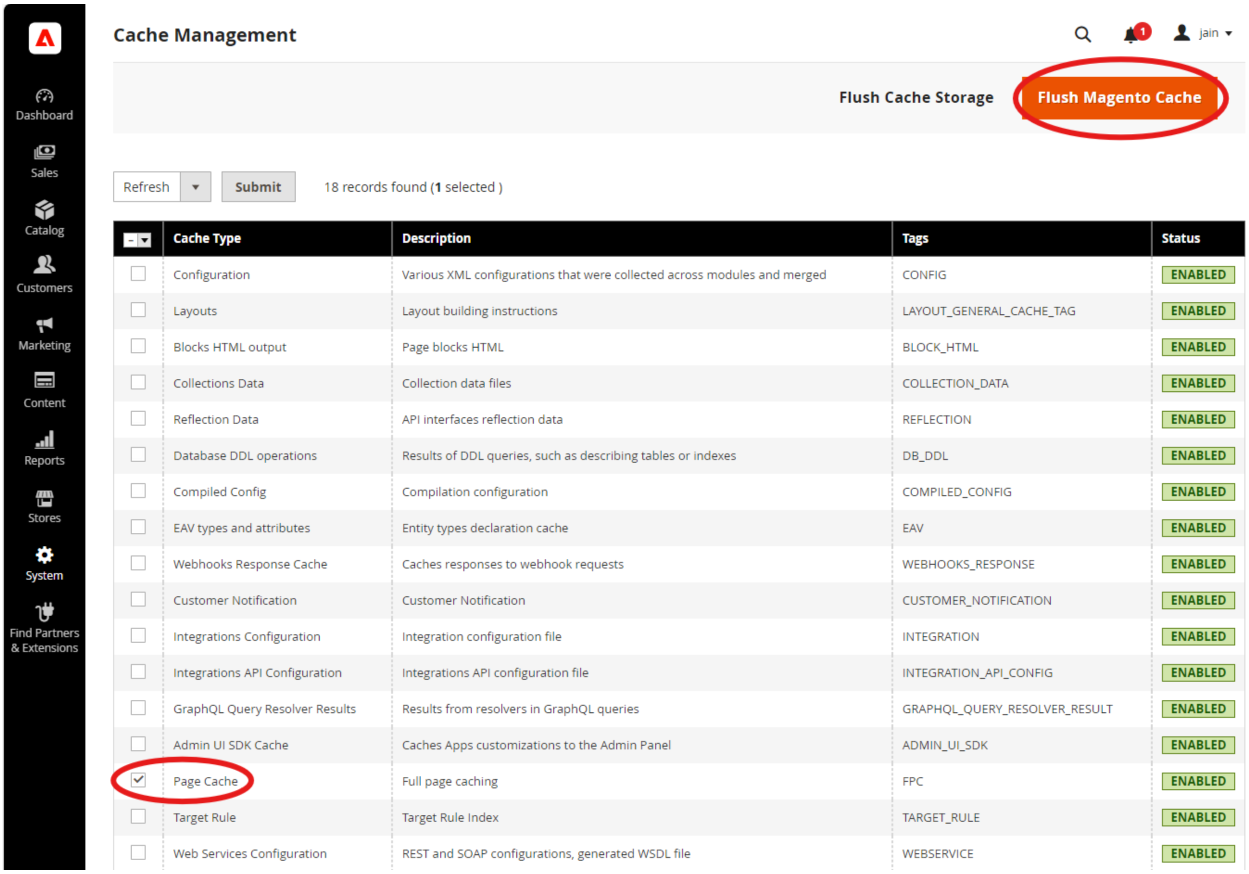
Upon saving, Adobe Commerce requires one to refresh the configuration by flushing the cache. To do this, you must do the following:

Navigate to Cache Management

System > Cache Management



Once within the Cache Management console, choose “Page Cache” and then “Flush Magento Cache”.



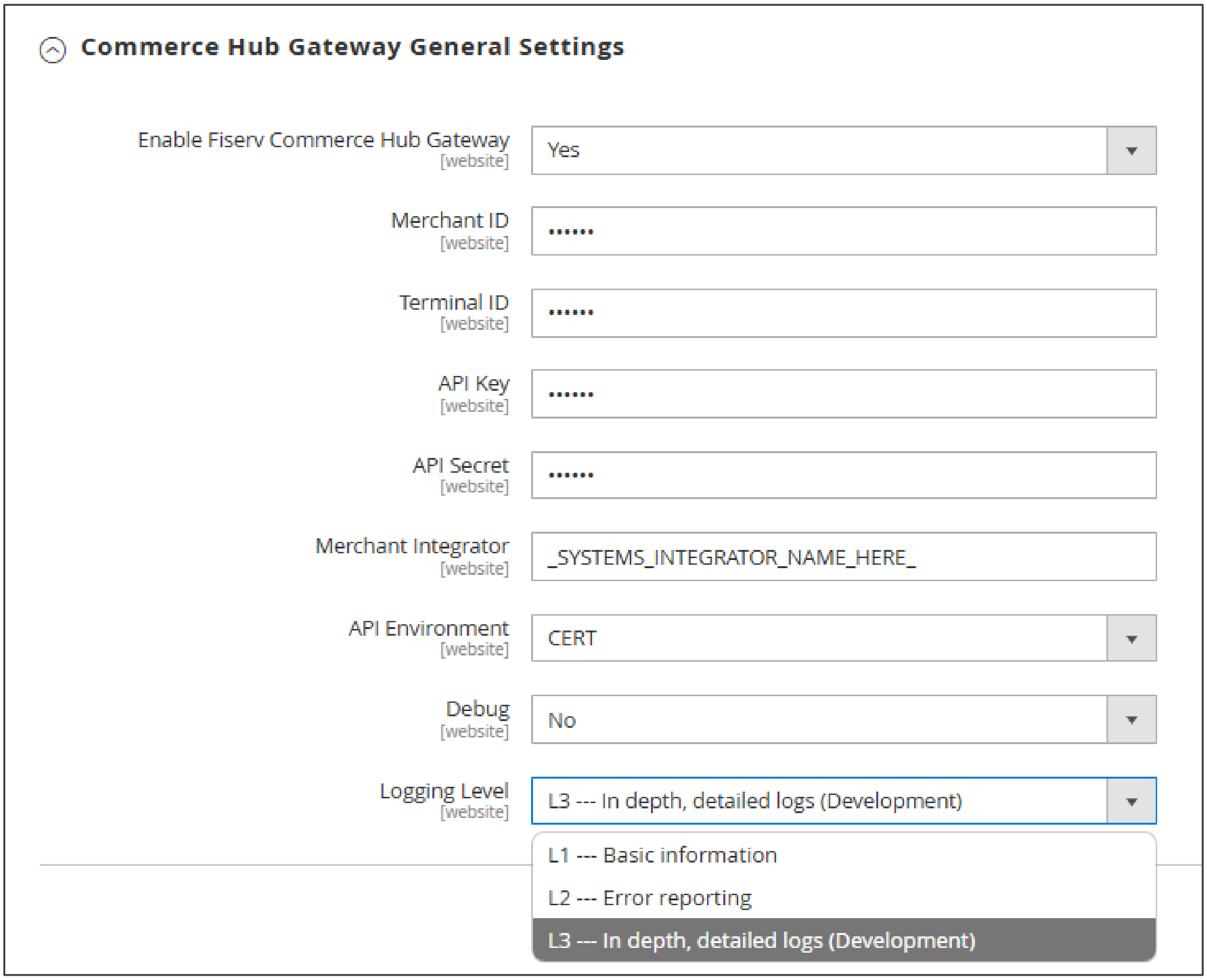
**<u>NOTE</u>: Each time a configuration setting is saved, the cache must be flushed for the changes to take effect.

Step 3: Configuration Instructions for Fiserv Payments

This section explains the various configuration areas, functions, and purpose.

Commerce Hub Gateway General Settings

This area determines the setup required to run transactions with the Adobe extension.



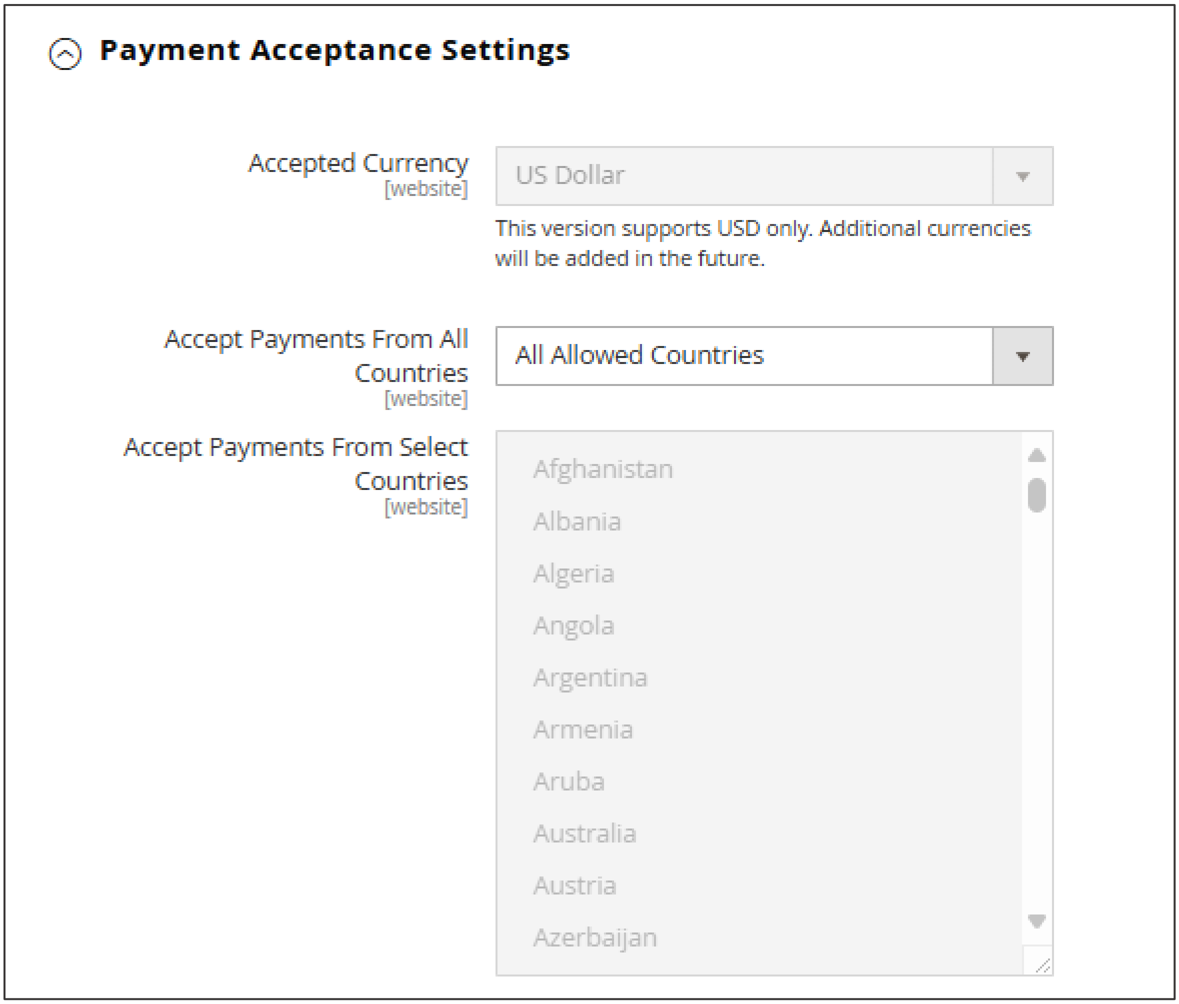
Configuration Field	Value	Description
Enable Fiserv Commerce Hub Gateway	Boolean: Yes / No	Setting this to Yes enables the use of the extension.
Merchant ID	String	Provided through Dev Studio.

Terminal ID	String	Provided through Dev Studio.
API Key	String	Provided through Dev Studio.
API Secret	String	Provided through Dev Studio. **NOTE. Please be mindful that there are different credentials and identifiers associated with the different environments. Please ensure you have the correct ones for both CERT and PROD, respectively.
Merchant Integrator	String	Please enter the entity that integrated the Adobe extension on behalf of the merchant – could be the merchant themselves or a 3rd party systems integrator.
API Environment	Values: • CERT • PROD	Fiserv supports two environments – CERT and PROD. **CERT** is the Carat development and testing environment for merchants and integrators. This is the environment where merchants and integrators will set up, configure, and perform UAT (user acceptance testing) for their end-to-end validation of how their web storefront will interact with the plugin. It is the merchant's responsibility to ensure their workflows and business rules are met. **PROD** is the Carat production environment. Once validation is complete, this is how a merchant implementation of the plugin can be promoted to a production environment for LIVE transactions.
Debug	Boolean: Yes / No	This configuration previously enabled and disabled logging of the plugin. This has since been deprecated and replaced by the Logging Level configuration.
Logging Level	Values: • L1 – Basic information	See the Transaction Logging section to learn more.

	• L2 – Error reporting • L3 – In depth, detailed logs (Development)	

Payment Acceptance Settings

This area determines the general payment acceptance in the extension.

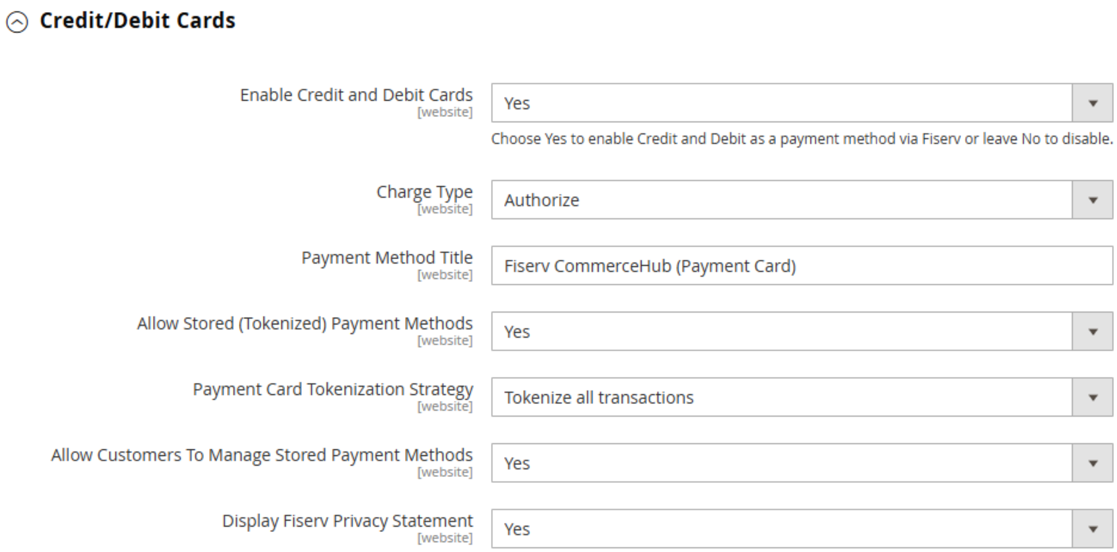


Configuration Field	Value	Description
Accepted Currency	US Dollar	US Dollar (USD) Only **NOTE. In this version of the plugin, only US Dollar (USD) is enabled. More currencies will be available over time.

Accept Payments From All Countries	Values: • All Allowed Countries • Specific Countries	This configuration determines from which countries consumers can purchase from the merchant website. This is typically based on merchant business rules and their operational ability to service consumers. If "**All Allowed Countries**" is chosen, a merchant can accept payments from all countries supported by Fiserv. All Allowed Countries list is the list countries that Fiserv can support--which considers geo-political and sanction mandates. If "**Specific Countries**" is chosen the following configuration field "**Accept Payments From Select Countries**" will be enabled. Else, it will be grayed out and immutable.
Accept Payments From Select Countries	Dropdown List	This configuration is enabled when "**Accept Payments from Select Countries**" is chosen as the value of the previous field. One to many may be selected. At least ONE country must be chosen, else the integration will not work.

Credit/Debit Cards

This section defines payment card specific settings.



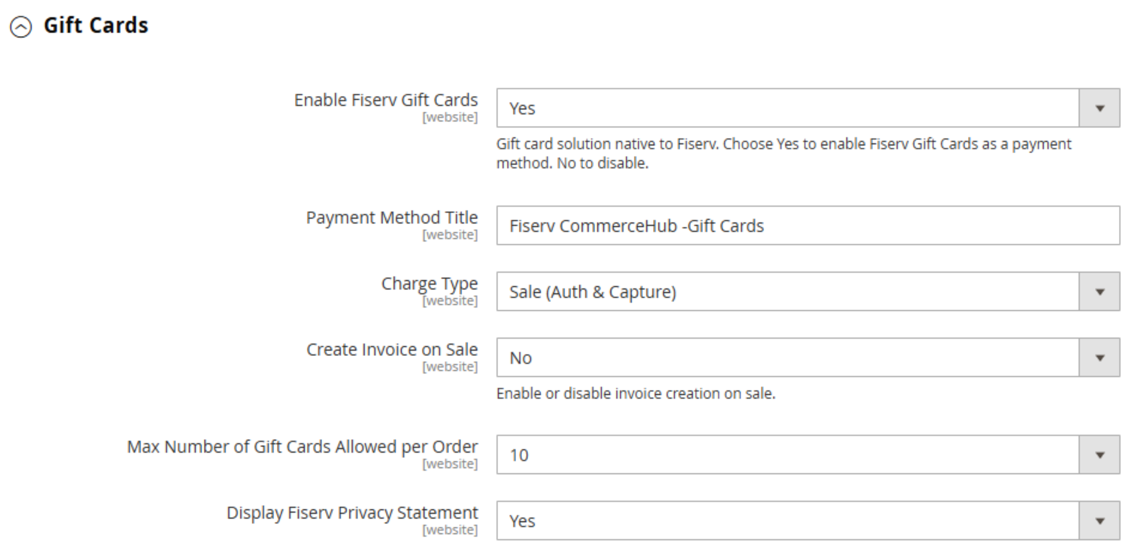
Configuration Field	Value	Description
Enable Credit and Debit Cards	Boolean: Yes / No	This configuration will enable the acceptance for credit and debit card features, and can be used independently of any other payment method in the Adobe extension. Yes, means that acceptance of credit and debit cards are enabled. No, means that the acceptance of credit and debit cards are disabled.
Charge Type	Values: • Sale (Authorization + Capture) • Authorize	Sale (Authorize + Capture) will fully charge the consumer at the time of checkout. Authorize requires a subsequent capture request. This charge type will place a hold / reserve on funds.

Payment Method Title	String	This is the label shown to the consumer for the credit / debit card section of available payment methods.
Allow Stored (Tokenized) Payment Methods	Boolean: Yes / No	This configuration will allow consumers to opt-in or opt-out to store (e,g, tokenize) their credit/debit cards within their merchant login profile. For future purchases when logged in, consumers can select a stored card on file rather than entering a new card. <u>Yes</u>, means that a check box will be shown at checkout so that a consumer can opt to save their card details; **DEFAULTED to Yes** <u>No</u>, means that a check box will NOT be shown at checkout. A consumer card does not have the option of being saved.
Payment Card Tokenization Strategy	Values: • Tokenize all transactions • Tokenize at customer request	This configuration is only enabled when **"Allow Customer To Store (Tokenize) Payment Methods"** is set to **Yes**. This configuration tokenizes a payment card 100% of the time or only when a consumer requests it. **Tokenize all transactions** enables the Adobe extension to tokenize all credit / debit card transactions. This is typically used when a merchant required a tokenized form of payment for downstream operational and customer service issues. **Tokenize at customer request only** tokenizes credit / debit cards when the consumer opts-in to save their form of payment. **NOTE. Consumers will only see their stored payment methods in their profile when the opt-in to save their form of payment at the time of purchase.

Allow Customers To Manage Stored Payment Methods	Boolean: Yes / No	This configuration is only enabled when **“Allow Customer To Store (Tokenize) Payment Methods”** is set to **Yes**. This configuration is tied to the above value. If "Allow Customers To Store (Tokenize) Payment Methods" = No, then this configuration field is not an option and will not even shown. If available, <u>Yes</u>, means that a consumer can log into their merchant store account, navigate to their stored payment method page, and add or delete a payment method (inside of their consumer profile). <u>No</u>, means that a consumer CANNOT log into their merchant store account and save or delete a stored payment method.
Display Fiserv Privacy Statement	Boolean: Yes / No	Displays a link to Fiserv’s Privacy notice within the checkout for this payment method <u>Yes,</u> means display Fiserv privacy statement <u>No,</u> means hide Fiserv privacy statement

Fiserv Gift Cards

This section defines Fiserv Gift Cards specific settings.



Configuration Field	Value	Description
Enable Fiserv Gift Cards	Boolean: Yes / No	This configuration enables or disables gift card acceptance. Yes, enables gift card acceptance. No, disables gift card acceptance. **NOTE. This feature requires an additional agreement / addendum for enablement. Please refer to your sales associate or relationship manager.
Payment Method Title	String	This is the label shown to the consumer for the gift card section of available payment methods.
Charge Type	Values: • Sale (Authorization + Capture) • Authorize	Sale (Authorize + Capture) will fully charge the consumer at the time of checkout. Authorize requires a subsequent capture request. This charge type will place a hold / reserve on funds.

Create Invoice on Sale	Boolean: Yes / No	Merchants can choose to generate invoices upon authorization or fulfillment for those orders that are covered entirely by gift card transactions.
Max Number of Gift Cards Allowed per Order	10	Up to 10 gift cards are allowed per order; Defaulted to “10”
Display Fiserv Privacy Statement	Boolean: Yes / No	Displays a link to Fiserv’s Privacy notice within the checkout for this payment method Yes, means display Fiserv privacy statement No, means hide Fiserv privacy statement

Step 4: Checkout Hosted Field Customizations

To ensure that we protect consumer payment information and help merchants manage their PCI and PII obligations, we have implemented a means to place all payment input data fields into Hosted Fields.

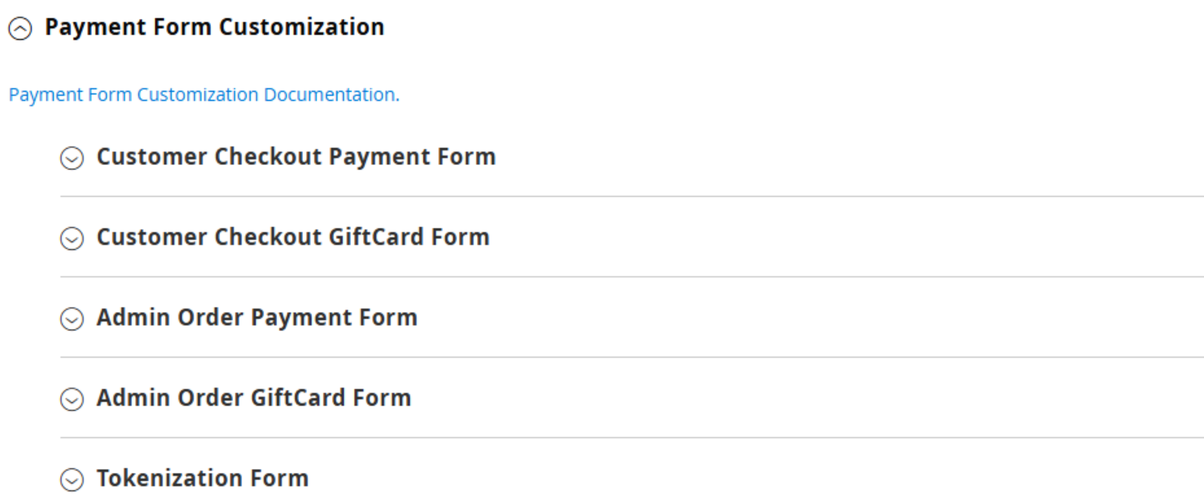
While we have helped manage the capture and input of sensitive payment data at checkout, we do recognize the importance of a consumer's experience with a merchant's brand and therefore have provided individual field customization options.

Customization Architecture

In the Adobe Commerce extension, there are a handful of touch points where consumer payment data can be entered – namely payment cards and gift cards.

- At Consumer Checkout
- Within the Admin Order interface
- Within the consumer's account profile

We have provided the following customization areas for management control.



Customizations leverage Checkout Hosted Fields. Please follow this link to view documentation.

https://developer.fiserv.com/product/CommerceHub/docs/?path=docs/Online-Mobile-Digital/Checkout/Hosted-Fields/Hosted-Fields.md&branch=main

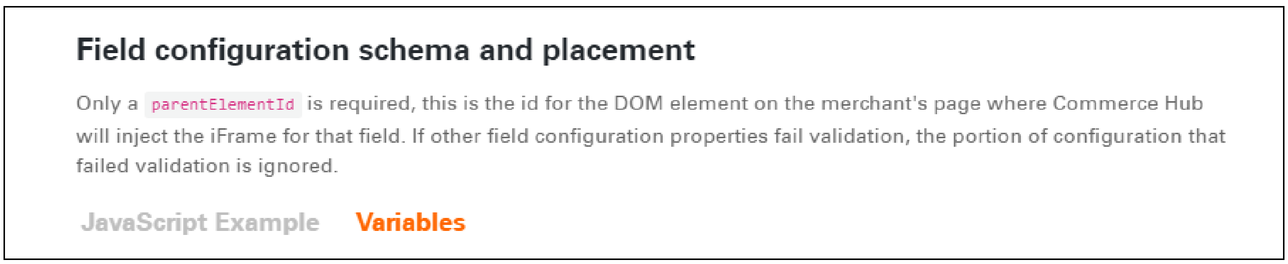
Field Customization

For the Adobe extension, we have taken away the complexity and nuances of field customizations by translating this into an easy-to-use interface.

Integration has been provided by the following supported fields:

```
cardNumber
nameOnCard
securityCode
expirationMonth
expirationYear
```

With the following field configurations which accommodates both payment cards and Fiserv gift cards (choose Variables tab):

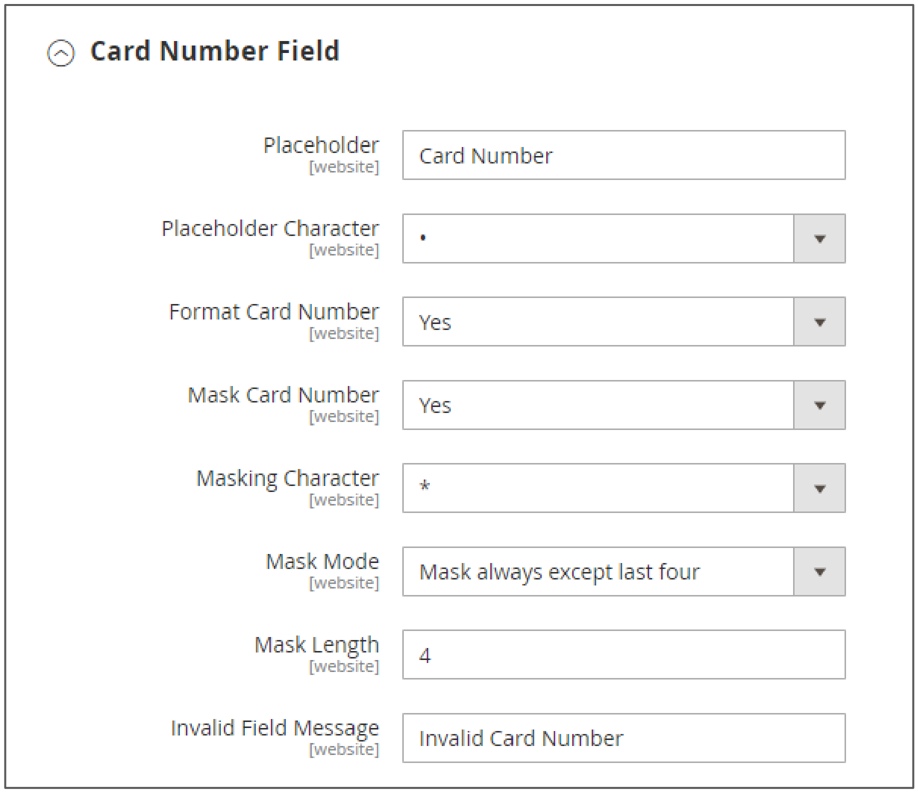



```
placeholder
dynamicPlaceholderCharacter
enableFormatting
masking.charcacter
masking.mode
masking.shrunkLength
```

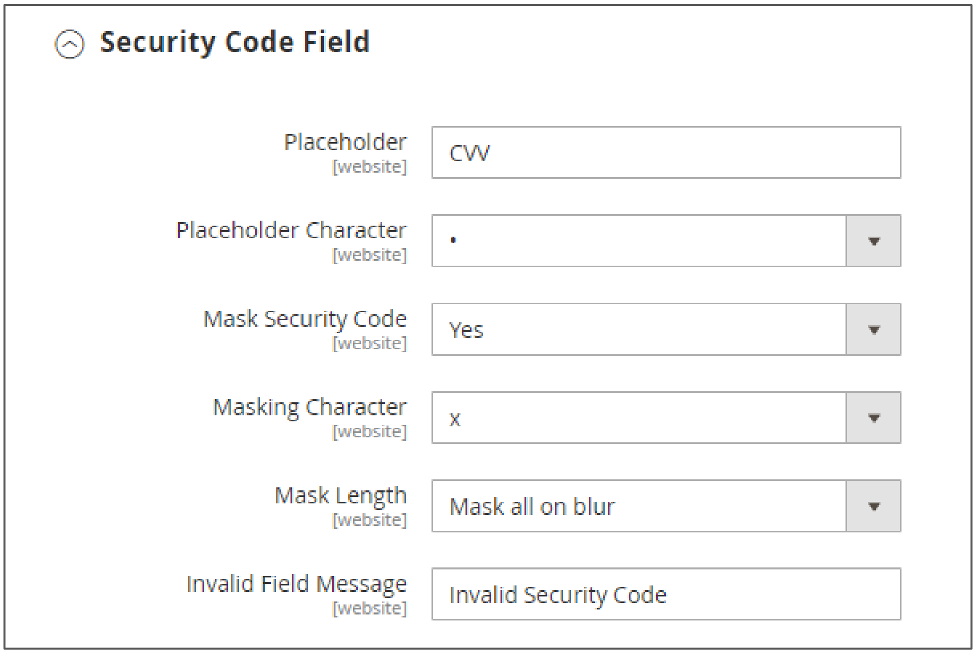
On the same page, see masking modes that are available for those customarily sensitive data fields.

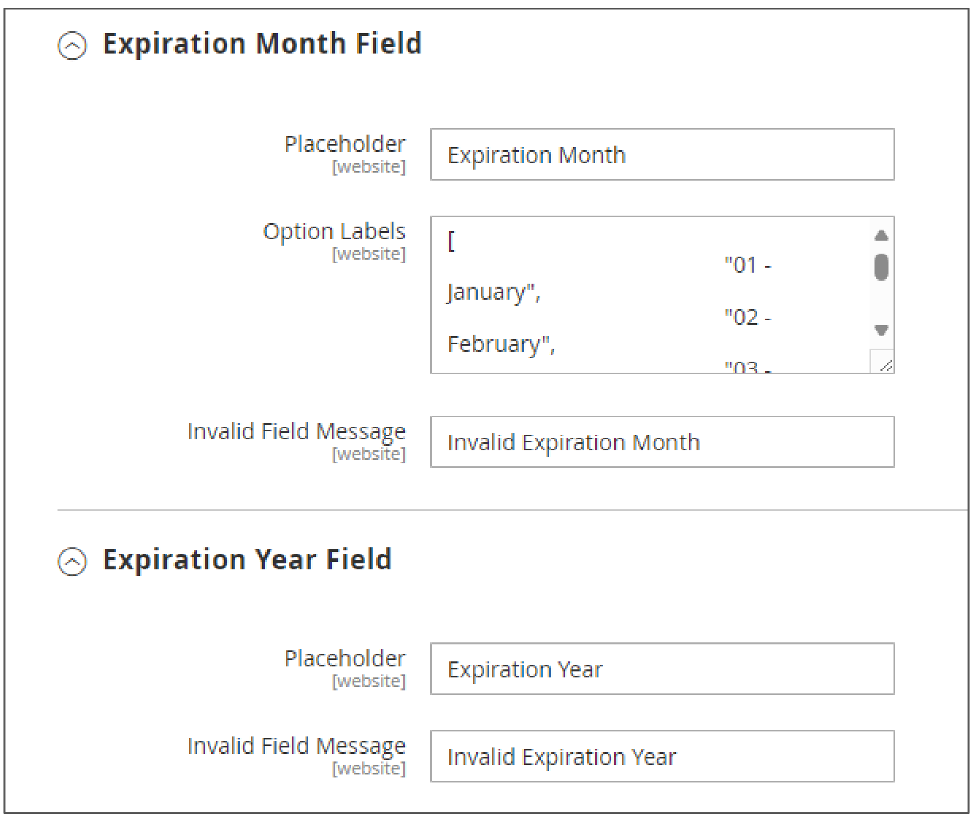
Each supported field will differ. For example, treatment of a credit card field is different than that of an expiration year field. Each field has been provided with options fit for purpose.

See below screenshots of each available field and options.

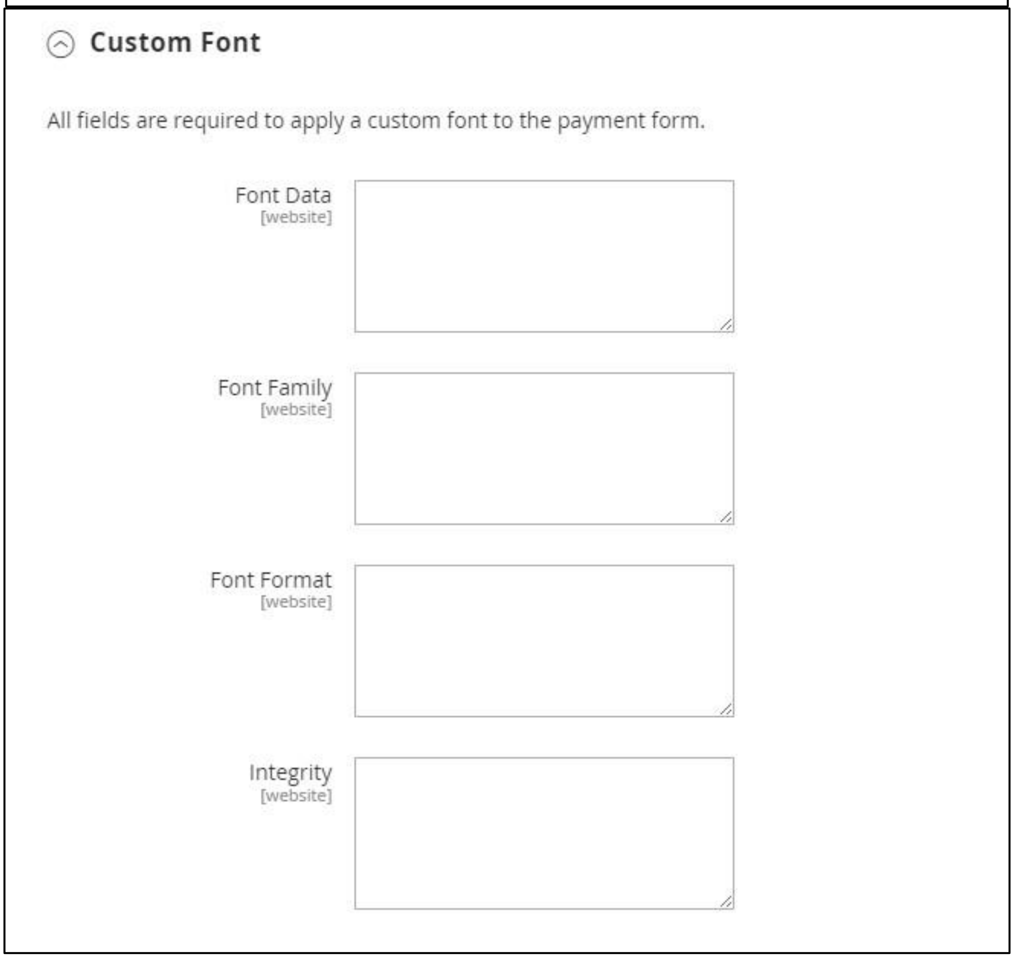

Card Number Field
Placeholder
[website]
Card Number
Placeholder Character
[website]
•
Format Card Number
[website]
Yes
Mask Card Number
[website]
Yes
Masking Character
[website]
*
Mask Mode
[website]
Mask always except last four
Mask Length
[website]
4
Invalid Field Message
[website]
Invalid Card Number


Name on Card Field
Placeholder
[website]
Name on Card
Invalid Field Message
[website]
Invalid Name on Card


Security Code Field
Placeholder
[website]
CVV
Placeholder Character
[website]
•
Mask Security Code
[website]
Yes
Masking Character
[website]
x
Mask Length
[website]
Mask all on blur
Invalid Field Message
[website]
Invalid Security Code


Expiration Month Field
Placeholder
[website]
Expiration Month
Option Labels
[website]
[
"01 -
January",
"02 -
February",
Invalid Field Message
[website]
Invalid Expiration Month
Expiration Year Field
Placeholder
[website]
Expiration Year
Invalid Field Message
[website]
Invalid Expiration Year


Form CSS
Checkout Form CSS
[website]
{
"input, select" : {
"background-color" : "#ffffff"
},
"input::placeholder": {
"color": "#000"
}
}


Custom Font
All fields are required to apply a custom font to the payment form.
Font Data
[website]
Font Family
[website]
Font Format
[website]
Integrity
[website]

Default Customization

For convenience, we have provided default customization. The intention is to provide an example of what can be done, and merchants can implement according to their brand requirements.

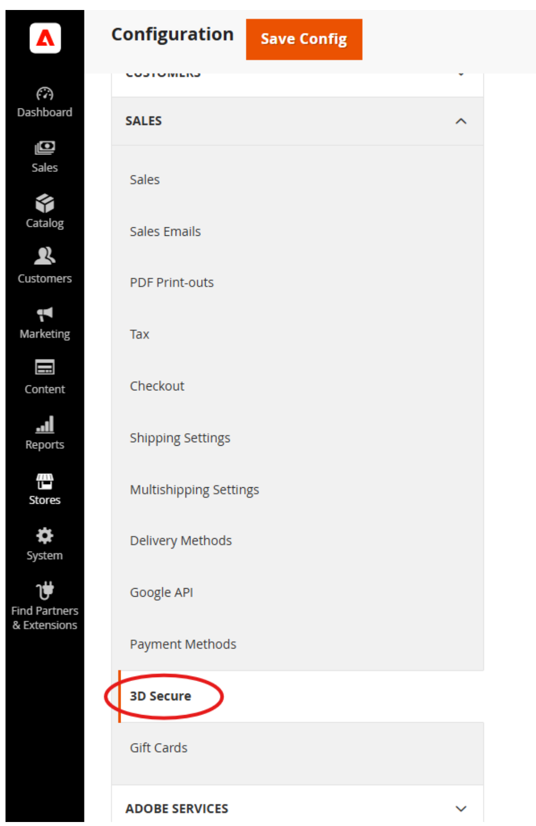
CSS values passed into the corresponding Form CSS fields will style inputs embedded within the secure card capture iFrames. To style surrounding DOM elements within the payment form template, override styles in the following stylesheets:

1. Customer checkout form: ch-form.css
2. Admin order creation payment form: ch-form-admin.css
3. Stored payment accounts form: ch-form-token.css

Step 5: 3D Secure Configuration

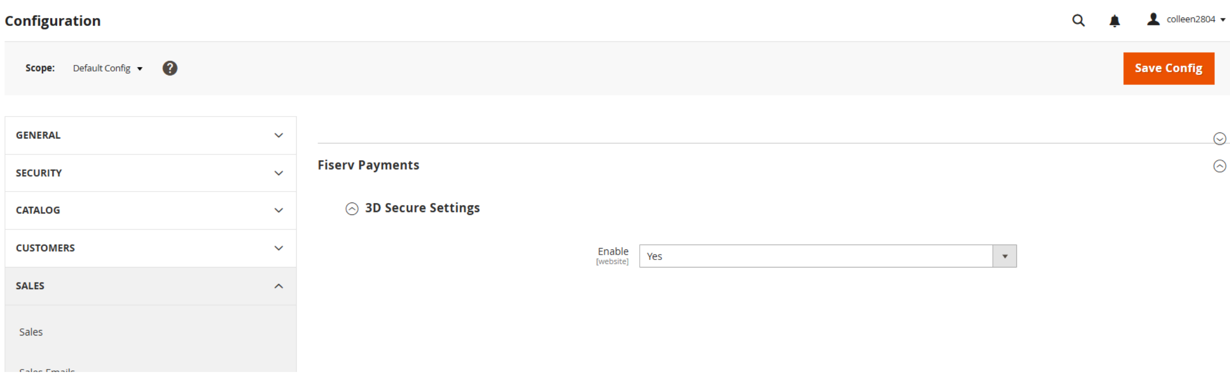
Navigate to 3D Secure

Stores > Configuration > Sales > 3D Secure



Expand 3D Secure Settings

Stores > Configuration > Sales > 3D Secure > Fiserv Payments > 3D Secure Settings



Configuration Field	Value	Description
Enable	Boolean: Yes / No	This configuration enables or disables 3D Secure functionality Yes, enables 3D Secure No, disables 3D Secure **NOTE. This feature requires an additional configuration for enablement. Please refer to your sales associate or relationship manager.

User Acceptance Testing (UAT)

Fiserv has tested and regressed our Adobe extension and feels that it is good practice to share relevant test scripts from a payment point of view with our extension users.

Payment Test Scripts

Please refer to the accompanying test scripts provided along with this document. We strongly recommend that merchants use these to complement their own testing processes.

End to End Testing

While the Adobe Plugin from Fiserv has been tested and regressed, it is the responsibility of the merchant to validate that their business rules are met pre-checkout, during checkout, and post-checkout.

This includes, but not limited to validating workflows, data requirements, and successful integration with upstream and downstream systems and tools leveraged by the merchant outside of core Adobe Commerce.

As part of end to end validation, please ensure that the Adobe administrative interfaces and Control Center from Fiserv (reporting system) meet the merchant's needs.

Operations

Timeout Handling

The Fiserv Adobe Extension has built-in management to handle timeout and latency issues between the Adobe extension (client) and Fiserv (server) -- covering a range of scenarios:

- Issues with requests between the client and server
- Issues with responses between the server and client
- Issues encountered during server-side processing

In principle, the timeout logic attempts to stay true to the transaction status on the server side. The Fiserv system is set to timeout at 20 seconds and the extension is configured to timeout at 22 seconds – accounting for round trip time.

Within the merchant's MID configuration, a custom timeout of 20 seconds is required. Please consult with your relationship manager / account manager to ensure this is set up properly.

To provide visibility on how timeout scenarios have been addressed, we employ the following steps.

Step 1) Idempotency. If a transaction initiated by the Adobe Extension (client) reaches the timeout, the extension will automatically retry the transaction leveraging Commerce Hub's idempotency feature.

- If successful, the client will receive the proper result of the payment request and the customary workflow will resume.
- If unsuccessful, perform Step 2.

Step 2) If idempotency is unsuccessful, the Adobe Extension (client) will perform a Transaction Inquiry request in an attempt to confirm the status of the original transaction.

- If successful, the client will be returned the status and confirmation of the original payment request. Merchant workflow will resume.
- If unsuccessful, perform Step 3.

Step 3) If the Transaction Inquiry is unsuccessful, the Adobe Extension then goes into recovery mode by doing the following:

- For authorization requests, we will Cancel the transaction to release any funds that were potentially reserved on the consumer's form of payment.
- For cancels, refunds, captures, and sale transactions, it is difficult for us to reverse as we do not want to attempt to reverse any money moving events. We suggest merchant operations investigate and take appropriate action with their consumers.

All timeout management steps above are logged and available for review. We suggest that there should be monitoring of these events by merchant operations.

Transaction Logging

System and transaction logs are critical for investigation and troubleshooting. Fiserv endeavors to provide operators with relevant and easy-to-use tools.

Transaction Log Enablement- Logging is automatically enabled defaulted to L2. This is so that a user can see both successful and errored transactions out-of-the-box.

Transaction Log Timezone – The default timezone for transaction logging is based on the Timezone set within Adobe Commerce's native framework. Below are the steps to change the timezone:

1. **Within the Adobe Admin Dashboard, navigate to Configuration**:
    a. From the Admin sidebar, go to `Stores > Configuration.`
2. **Select General Configuration**:
    a. Under `General`, expand the `Locale Options` section.
3. **Change Timezone**:
    a. Locate the `Timezone` dropdown and select the desired timezone.
4. **Save Configuration**:
    a. Click on `Save Config` to apply the changes.

Transaction Log Location – Logs are located by default within the logging architecture of Adobe Commerce into a dedicated log file. Fiserv Payments transaction logs can be found at:

```
<Adobe Commerce Root Directory>/var/log/commerce_hub.log
```

Logging Levels – The available logging levels have been designed to provide an increasing level of verbosity to assist operators in research and troubleshooting. There are a total of three logging levels.

L1 represents the base log level that tracks the active process within the plugin.

L2 reports errors. It will include both error messages and stack traces.

L3 includes the most verbose information that includes information of objects created within the plugin throughout the transaction lifecycle.

At installation, logging is automatically set at L2, but can be changed without having to restart the Adobe Commerce instance; however, the cache must be flushed for the new configurations to take effect.

When the logging level is increased (e.g. from L2 to L3), lower-level logs will still be enabled and present. If a log level is decreased (e.g. from L3 to L1), log information appropriate to the new log level will be written.

L1 Log Example:

```
[2024-10-09T01:57:53.855833+00:00] CommerceHubLogger.INFO: [L1] Initiating Credentials Request [] []
[2024-10-09T01:57:54.612885+00:00] CommerceHubLogger.INFO: [L1] Credentials request success [] []
[2024-10-09T01:58:55.773643+00:00] CommerceHubLogger.INFO: [L1] Retrieving payment tokens [] []
[2024-10-09T01:58:55.774269+00:00] CommerceHubLogger.INFO: [L1] Payment tokens retrieved [] []
[2024-10-09T01:58:56.999798+00:00] CommerceHubLogger.INFO: [L1] starting execution [] []
[2024-10-09T01:58:57.004390+00:00] CommerceHubLogger.INFO: [L1] Initiating Token Auth Transaction [] [] . . .
[2024-10-09T10:19:39.443676+00:00] CommerceHubLogger.INFO: [L1] Transaction success [] []
[2024-10-09T10:19:39.443726+00:00] CommerceHubLogger.INFO: [L1] Transaction ID:
a91e532c15104b30b930f3e9caee4c12 [] []
[2024-10-09T10:19:39.443726+00:00] CommerceHubLogger.INFO: [L1] Transaction ID:
a91e532c15104b30b930f3e9caee4c12 [] []
```

L2 Log Example:

```
[2024-10-09T12:58:59.249871+00:00] CommerceHubLogger.ERROR: [L1] Transaction failure. Commerce Hub response
returned with unsuccessful transaction state [] []
[2024-10-09T12:58:59.249922+00:00] CommerceHubLogger.ERROR: [L1] Transaction ID:
4ea07cf0e13e4dacaa5ad5264106fc2c [] []
[2024-10-09T12:58:59.249970+00:00] CommerceHubLogger.ERROR: [L2] Transaction state: DECLINED [] []
[2024-10-09T12:58:59.250014+00:00] CommerceHubLogger.ERROR: [L2] Response message: Declined [] []
[2024-10-09T12:58:59.250057+00:00] CommerceHubLogger.ERROR: [L2] Payment Source Type: PaymentToken [] []
[2024-10-09T12:58:59.285476+00:00] CommerceHubLogger.INFO: [L1] Processing ValueLink transactions. Total
transactions: 1 [] []
[2024-10-09T12:58:59.285558+00:00] CommerceHubLogger.INFO: [L1] Attempting to cancel ValueLink transaction ID:
0653fd60b4a14e56adb4ebff0c0a7de4 [] []
[2024-10-09T12:58:59.689786+00:00] CommerceHubLogger.INFO: [L1] Successfully canceled ValueLink transaction ID:
0653fd60b4a14e56adb4ebff0c0a7de4 [] []
[2024-10-09T12:58:59.689888+00:00] CommerceHubLogger.INFO: [L1] All ValueLink transactions were successfully
voided. [] []
L3 Log Example:
```

L3 Log Example:

```
[2024-10-09T01:39:24.739971+00:00] CommerceHubLogger.INFO: [L3] Response Body:
```


```
{
  "gatewayResponse": {
    "transactionType": "CHARGE",
"transactionState": "AUTHORIZED",
"transactionOrigin": "ECOM",
"transactionProcessingDetails": {
  "orderId": "CHG014d76b82860de691755fce82d46ace528",
  "transactionTimestamp":
  "2024-10-09T01:39:22.957077478Z",
  "apiTraceId": "50b5bff73eeb441992a1ac7dca6a6f7e",
  "clientRequestId": "1728987324033",
  "transactionId": "50b5bff73eeb441992a1ac7dca6a6f7e",
  "apiKey": "AkfAAfn6jtD0wNn5luIPHGftBSgPeW3B"
  }
},
"paymentReceipt": {
"approvedAmount": {
  "total": 25,
  "currency": "USD"
},
"processorResponseDetails": {
"approvalStatus": "APPROVED",
"approvalCode": "OK5783",
"referenceNumber":
"ac7dca6a6f7e", "processor":
"FISERV", "host": "NASHVILLE",
"networkRouted": "VISA",
"networkInternationalId": "0001",
"responseCode": "000",
"responseMessage": "Approved",
"hostResponseCode": "00",
"hostResponseMessage": "APPROVAL",
"additionalInfo": [
   { "name": "COUNTRY_CODE",
   "value": "USA"
 },
 {
   "name": "CARD_PRODUCT_ID",
   "value": "X"
```

```
 },
 {
   },
   "networkDetails": {
     "network": {
   "network": "Visa" },
   "networkResponseCode": "00",
   "cardLevelResultCode": "H",
   "validationCode": "CN ",
   "transactionIdentifier": "014283786163273"
 },
 "cardDetails": {
   "binSource": "FISERV",
   "recordType": "DETAIL",
   "lowBin": "4111111111",
   "highBin": "4111111111",
   "binLength": "10",
   "binDetailPan": "16",
   "countryCode": "USA",
   "detailedCardProduct": "VISA",
   "detailedCardIndicator": "DEBIT",
   "pinSignatureCapability": "SIGNATURE",
   "issuerUpdateYear": "22",
   "issuerUpdateMonth": "04",
   "issuerUpdateDay": "22",
   "regulatorIndicator": "REGULATED",
   "cardClass": "CONSUMER",
   "anonymousPrepaidIndicator": "NON_ANONYMOUS",
   "productId": "MCS",
   "accountFundSource": "DEBIT",
   "panLengthMin": "16",
   "panLengthMax": "16",
   "moneySendIndicator": "UNKNOWN"
 },
 "source": {
   "sourceType": "PaymentToken",
   "tokenData": "1079847268701111",
   "card": {
     "expirationMonth": "01",
 "expirationYear": "2025",
 "bin": "411111",
 "last4": "1111",
 "scheme": "VISA"
},
"accountType": "FPAN"
  }
} [] []
[2024-10-09T01:58:58.877531+00:00] CommerceHubLogger.INFO: [L1] Transaction success [] []
[2024-10-09T01:58:58.877581+00:00] CommerceHubLogger.INFO: [L1] Transaction ID:
e7f4a975ca574ba891b9473a5de19fb8 [] []
```

Known Issues / Questions

- **Form Validations.**
  - For reasons flagged for ADA for those who are visually impaired, we have taken the liberty to produce a readable error that indicates a potential input error and asks for the information to be re-entered.
  - Iframe border validations can persist after reloading the form
  - Error banner for tokenization form can disappear after only a second of visibility
  - Iframe width on mobile screens can be incorrect

- **Decline Messages.** Where transactions result in soft and/or hard declines, the extension will report the reasons within the response object. It is fully available for a merchant to access, interpret, and render an appropriate message for the end user. We assume that this is the case to provide full control to a merchant.

- **Admin refresh issue.** When a merchant sales associate is entering an order on behalf of a merchant in the Adobe Commerce Admin interface, there is an edge case where if the page is refreshed, the form fields will disappear. With research, this is not within the scope of the Fiserv extension, but more so with the Adobe platform. We plan on reporting this to Adobe.

- **2038 Problem.** Attempting to tokenize a card with an expiration date later than January 19, 2038 overflows the datatype, which causes it not to be displayed in the Magento UI. This is a bug in Magento's framework and needs to be addressed by Adobe.

- In select scenarios, when PayPal vaulting is enabled and an authenticated user updates their shipping address, the shipping address displayed in PayPal's checkout does not match the shipping address in Adobe Commerce. This will be addressed in a future release.

- Users can update the shipping address in the PayPal checkout modal. This change is not communicated back to Adobe Commerce and can result in a mismatch between the shipping address listed in a user's PayPal account and the address listed in Adobe Commerce. This will be addressed in a future release.

Releases Notes

Version 1.4.1

- Synchronize versioning

Version 1.3.2

- Bug fixes
  - Addressed change in Commerce Hub Balance Inquiry handling.
- Enhancement

- Upgraded Commerce Hub Checkout SDK to 3.7.16

Version 1.3.1

- Bug fixes
    - Resolved PHP CURL library header miscalculation.

Version 1.3.0

- Enhancements
    - Apple Pay for one-time payments has been enabled as a payment method through Fiserv Commerce Hub.
        - Apple Pay for recurring payments (subscriptions) will be a part an upcoming release
    - PayPal has been enabled as a payment method through Fiserv Commerce Hub.
        - Including PayPal vaulting for authenticated users.
    - PayPal Fastlane has been enabled as an alternative payment flow through Fiserv Commerce Hub.

Version 1.2.2

- Enhancements
    - Merchants can now run 3D-Secure transactions using saved payment methods (tokens). This enhancement ensures that tokenized payments benefit from the same level of security as traditional card payments.
    - Updates were made to the gift card balance inquiry function to support internal Commerce Hub changes.

Version 1.2.1

- Fixed
    - Compatibility with Adobe Commerce 2.4.8-p1 including:
        - Monolog LineFormatter format change.
        - Handle Braintree OrderAdapter `getBaseGrantTotal()` float bug.

Version 1.2.0

- Fixed
    - A “Clear” button has been added to the gift card balance check interface, allowing users to reset input fields.
- New Features
    - Unsuccessful Orders Dashboard
        - A new dashboard has been introduced to provide detailed, exportable insights into orders with unsuccessful transactions, including cause of failure, payment gateway response, and other relevant information.
    - 3D Secure Integration

        - Support for 3D Secure has been added to enhance payment card transaction security.
        - 3D Secure compatibility with vaulted payment methods is pending for a future release.
    - Partial Capture Support for Gift Cards
        - Merchants can now perform multiple partial captures on gift card transactions. Requires premium gift solutions entitlement.
    - Multiple Gift Card Payment Support
        - Customers can apply multiple gift cards to a single order, up to a configurable maximum (default: 10).
        - Includes split payment orchestration (e.g. orders split between gift and payment cards).
    - Configurable Invoice Creation for Gift Card Transactions
        - Merchants can choose to generate invoices for orders covered entirely by gift card transactions upon authorization or fulfilment.
    - Displaying Fiserv Privacy Statement
        - A link to Fiserv's Privacy Statement is now displayed under all Commerce Hub payment methods in the checkout.
        - Merchants can toggle visibility of the privacy statement in the configuration panel.

Version 1.1.0

- Fixed:
    - Bug where $0 capture was triggered for invoices completely covered by gift card or store credit balances
    - Bug where $0 refund was triggered for credit memo's completely covered by gift card or store credit balances
    - Bug where gift card line item appeared on invoices/credit memos without gift cards

Version 1.1.0

- Enhancements
    - Added new configuration named "Merchant Integrator" to mark the entity / 3rd party integration that integrated the Adobe extension. Meant to assist support incidents.
    - Credit / debit cards and gift cards can now be used independently of one another.
    - Credit / debit card tokenization option added to tokenize 100% of the time.
    - Minor Configuration UI changes.

Version 1.0.1

- Fixed.
    - Timeout reversal CURL header calculation
    - Sequencing of gift card captures

- Minor UI bugs

Version 1.0.0

- Fixed.
  - Propagated gift card transactions to email invoices.
  - Miscellaneous cosmetic fixes.
- New Feature
  - Timeout reversal management.
- Completed regressions and compatibility tests for production

Version 0.0.5

- Fixed. Updates and changes to .xml files and database schema for installation process.

Version 0.0.4

- New Features.
  - Gift card balance inquiry and redemption. Redemption includes both sale and authorization/capture functionality leveraging customizable field architecture.
  - Enhanced transaction logging with levels of verbosity.
  - Split tender to include a payment card, gift card, and store credit.

- Upgrade to Checkout SDK 3.1.9 for better browser and form factor support.

- Fixed. Various functional and regression defects.

Version 0.0.3

- Fixed. Reported that API Key and Secret fields were applied a validation check that resulted in a false negative. Remediated by modifying the validation logic.

- Fixed. Logic was implemented to message an end user when input data was invalid – e.g. bad formatting. Reported that certain field input checks that triggered "invalid data" messages were not being cleared once and end user had corrected the input. Input values were correct, yet an invalid message was still shown. While it did not prevent transactions from executing, it did provide usability issues and confusion for end users. Remediated the validation logic to clear these messages once corrected.

Version 0.0.2

- Fixed. Reported that certain Adobe Commerce installations do not allow system.xml files with nested objects greater than three levels preventing the plugin from starting. Remediated by collapsing the nested objects.

- Fixed. Reported that the checkout place order button was not able to handle parallel modules. Issue uncovered when the Avatax and Fiserv modules could not be executed without forcing a second click by the user. Remediated by changing the order of operations of how a parallel module is called.

- Updated configurations. The Fiserv Payments configuration page has been revamped for better categorization and usability – including ability to set invalid field messages. New configurations are reflected in this document.

Feedback

TBD